What you will learn in this chapter

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- You will discover that there is a hidden link between **information** and **complexity**.
- You will discover how computer science defines **information**, based exclusively on **coding** and computation.
- You will know why three mathematicians discovered the same notion of **complexity** at about the same time.
- You will learn how to estimate the complexity of passwords, of π, of lottery draws and of the cups in your grandmother's cupboard.

Marvin Minsky, one of the founding fathers of Artificial Intelligence,
gave **this great advice** about Algorithmic Information Theory:

"*Everybody should learn all about it and spend the rest of their lives working on it.*"

Algorithmic information is used in various sciences:

• data science	see Chapter 2
• mathematics	see Chapter 3
• machine learning	see Chapter 4
• cognitive sciences	see Chapter 5
• biology	see for instance: Vitanyi, P. & Cilibrasi, R. (2020). Phylogeny of the COVID-19 Virus SARS-CoV-2 by Compression. *bioRxiv*, 10.1101/2020.07.22.216242.
• physics	see for instance: Bennett, C. H. (1988). Logical depth and physical complexity. In R. Herken (Ed.), *The universal Turing machine - A half-century survey*, 227-257. Oxford: Oxford University Press.
• neurosciences	see for instance: Szczepański, J., Amigó, J. M., Wajnryb, E. & Sanchez-Vives, M. V. (2003). Application of Lempel--Ziv complexity to the analysis of neural discharges. *Network: Computation in Neural Systems, 14* (2), 335-350.
• linguistics	see for instance: Brighton, H. & Kirby, S. (2001). The survival of the smallest: stability conditions for the cultural evolution of compositional language. *LNAI Vol. 2159*, 592-601. Berlin: Springer Verlag.
• sociology	see for instance: Butts, C. T. (2001). The complexity of social networks: theoretical and empirical findings. *Social networks, 23* (1), 31-72.
• ecology	see for instance: Dakos, V. & Soler-Toscano, F. (2017). Measuring complexity to infer changes in the dynamics of ecological systems under stress. *Ecological Complexity, 32*, 144-155.
• ethology	see for instance: Ryabko, B. & Reznikova, Z. (2009). The use of ideas of information theory for studying "language" and intelligence in ants. *Entropy, 11* (), 836-853.

Conversely, it is hard to find a science that would be immune to its influence.
Algorithmic Information's ambition is to describe what it means for an intelligent entity, be it artificial or natural, to make sense of the world.

Detailed outline Chapter (1)

Describing data

Description length

We introduce the notion of description length, which is the length of a program used to generate an object.
The easiest way of describing objects is to distinguish them within a set.

Description complexity of integers

Integers can easily be distinguished by their rank in $\mathbb{N}$. But this method does not see why round numbers are simple.

Complexity and structure

Can we determine the complexity of an object by its structure? We illustrate this possibility by considering passwords: a good password should be *simple* to remember and *complex* to guess.

Defining complexity

We define Kolmogorov complexity, which is the core of this course. It corresponds to the minimal description length of an object. This quantity depends on a reference machine, but the "invariance theorem" makes it objective.

Conditional complexity

Sometimes, having access to another object can be a good proxy to provide an efficient description. This idea is involved in the definition of conditional complexity. A link between non-conditional complexity and conditional complexity is given by the "chain rule", which plays a prominent role, especially in AI.